

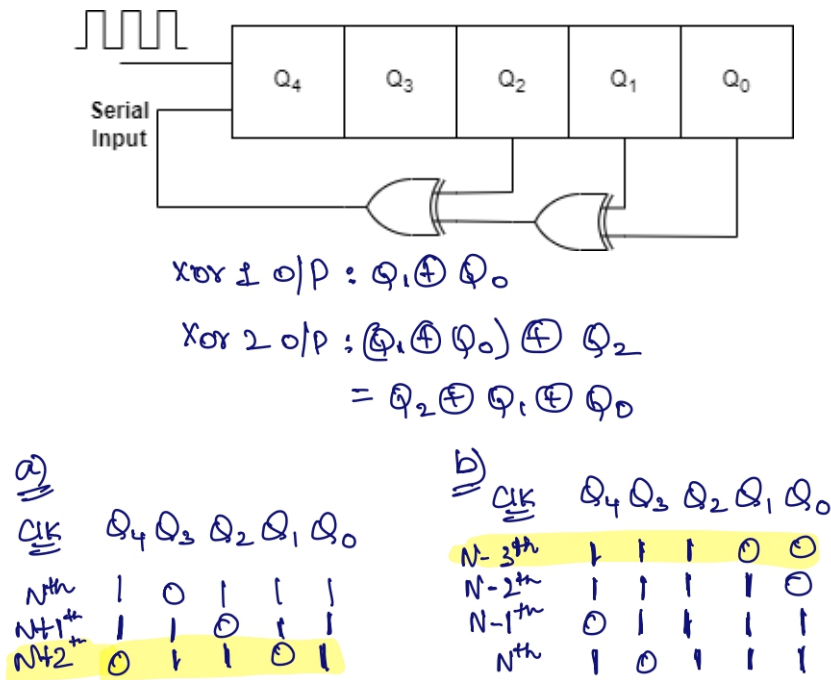
Digital Systems

Tutorial 7

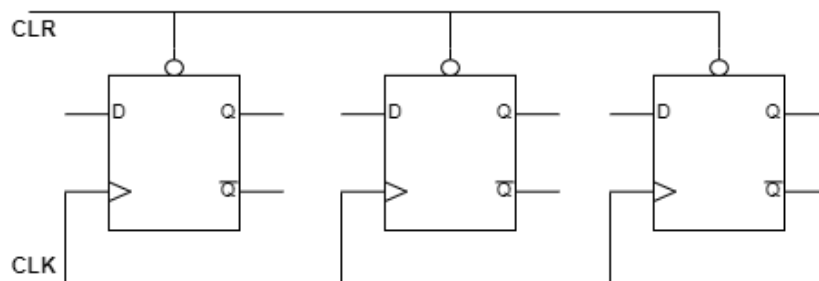
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1. Find the status of the following shift register: (i) after 2 clock cycles and (ii) 3 clock cycles prior to the present state. Present status of the register is $Q_4Q_3Q_2Q_1Q_0 = 10111$.



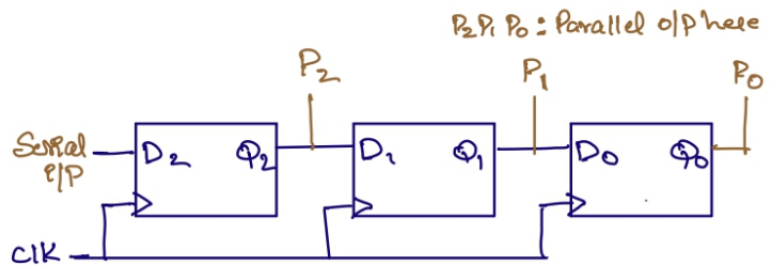
2. List out the outputs of each FF ($Q_2Q_1Q_0$) at the end of each clock period until all of the information is reached at the output, when it is used as a: i) SIPO register ii) SISO register iii) PISO register and iv) PIPO register. Assume the initial status of the register as 000 and data to be loaded is 101. Consider one time loading in the case of parallel loading.



How much time does it take for all the information given at the input to receive at the output if one clock period is T . Also generalize the time taken for n -bit registers for each case. For n -bit register take i/p information of n -bit length.

a) SIPO CLK $Q_2 Q_1 Q_0$

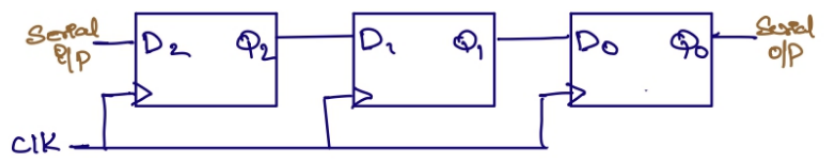
	000
1	100
2	010
3	101



Time: $3T$ (For 3-bit register)
 nT (For n -bit register)

b) SISO CLK $Q_2 Q_1 Q_0$

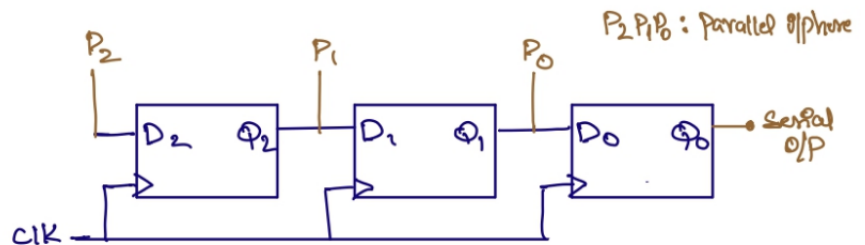
	000
1	100
2	010
3	101
4	-10
5	--1



Time: $5T$ (For 3-bit register)
 $nT + (n-1)T$ (For n -bit register)
 $= (2n-1)T$

c) PISO CLK $Q_2 Q_1 Q_0$

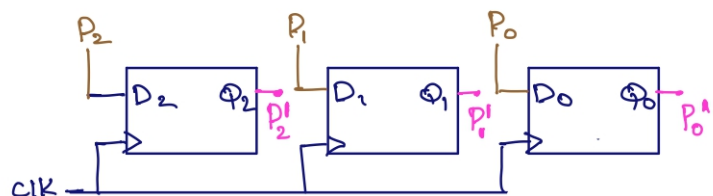
	000
1	101
2	-10
3	--1



Time: $3T$ (For 3-bit register)
 nT (For n -bit register)

d) PIPO CLK $Q_2 Q_1 Q_0$

	000
1	101



$P_2 P_1 P_0$: Parallel i/p

$P'_2 P'_1 P'_0$: Parallel o/p

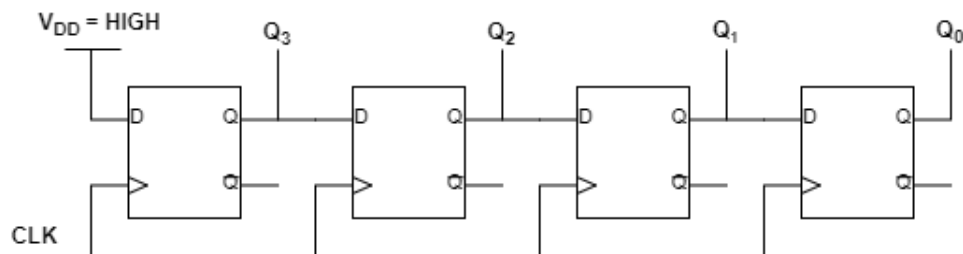
Time: T (For 3-bit register)
 T (For n -bit register)

3. The contents of a four-bit register are initially 0110. The register is shifted six times to the right, with the serial input being 1011100. What are the contents of the register after each shift?

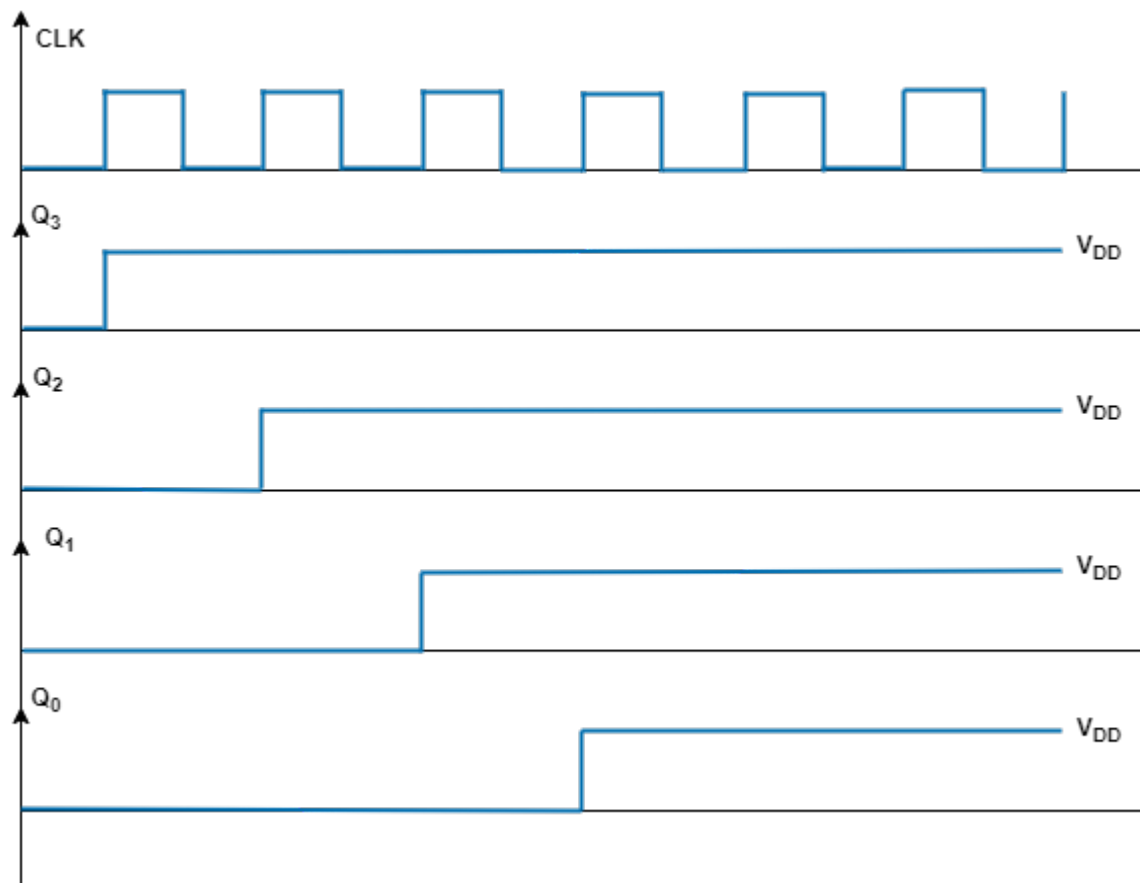
Initial content = 0110, serial i/p = 1011100

serial i/p	clk	Registers content
1011100	0	0 1 1 0
0	1	0 0 1 1
0	2	0 0 0 1
1	3	1 0 0 0
1	4	1 0 0 0
1	5	1 1 0 0
0	6	0 1 1 0
1	7	1 0 1 1

4. Complete the timing diagram for this circuit, assuming all Q outputs begin in the low state.



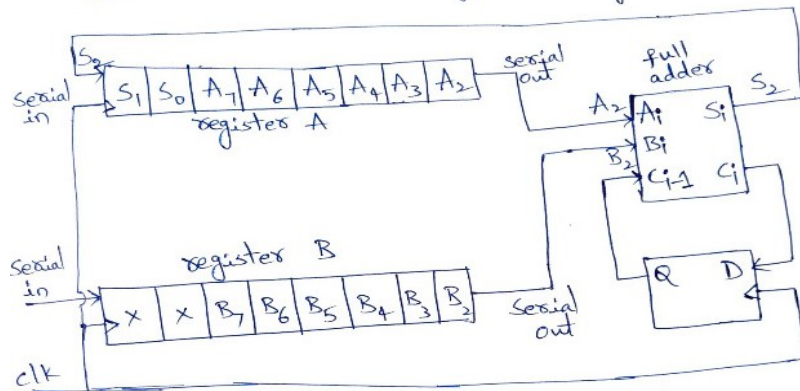
Write the content of the register after each shift with clock.



clk	Q_3	Q_2	Q_1	Q_0
—	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	1	1	1	1
6	1	1	1	1

5. Design an 8-bit serial adder using minimum number of shift registers.

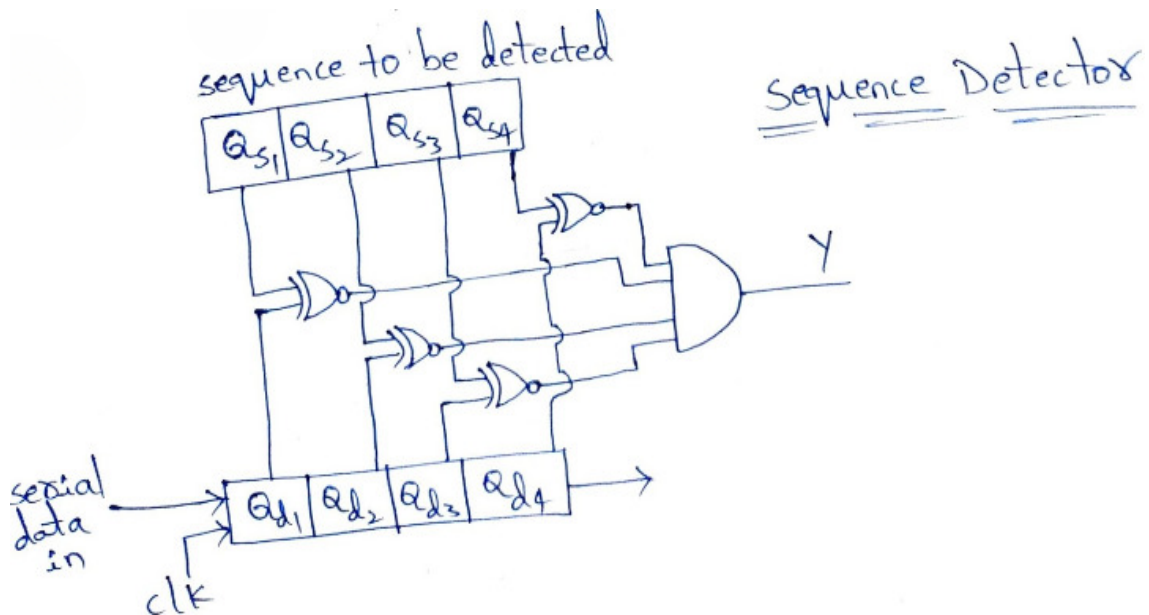
"One" full adder is enough to design n-bit serial adder



→ Minimally we can use 2 registers to design n-bit serial adder. The sum bits can be stored in any of the 2 registers to reduce the usage of more no. of memory elements.

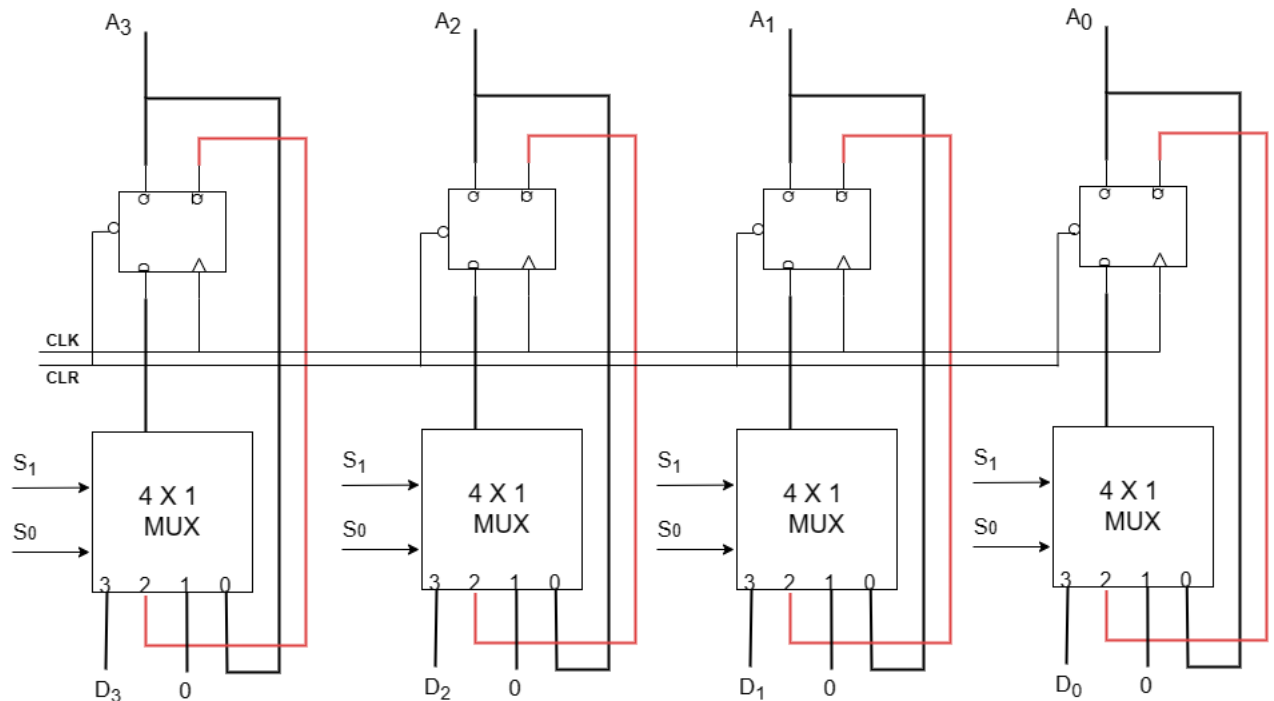
(design is shown from 2 clock cycles after the initialization)

6. Design a sequence detector using 4-bit shift registers.



7. Draw the logic diagram of a four-bit register with four D-flipflops and four 4×1 multiplexers with mode selection inputs s_1 and s_0 . The register operates according to the following function table.

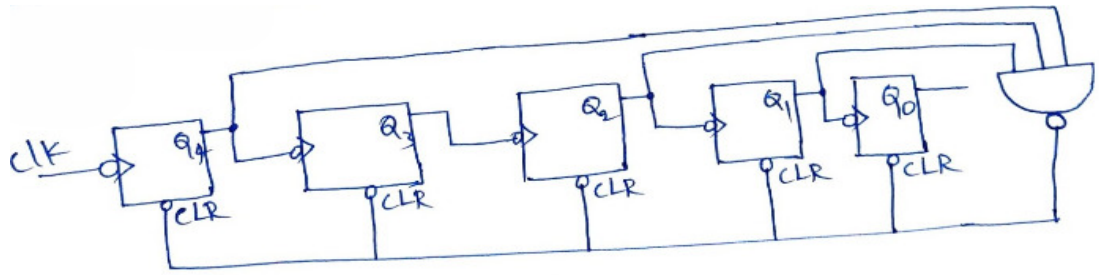
s_1	s_0	Register Operation
0	0	No change
1	0	Complement the four outputs
0	1	Clear register to 0 (synchronous with the clock)
1	1	Load parallel data



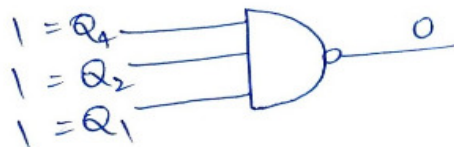
8. A traffic signal cycles from GREEN to YELLOW, YELLOW to RED, RED to GREEN. In each cycle, GREEN is turned ON for 90 seconds, YELLOW is turned ON for 5 seconds and RED is turned ON for 60 seconds. This traffic light system has to be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 seconds period. Find the number of flipflops required to implement the system. Implement this system using T-flipflops.

Total cycle time = $90 + 5 + 60 = 155 \text{ sec.}$
 clk period $t_{\text{clk}} = 5 \text{ sec.}$
 $\Rightarrow$ no. of clk cycles $n = \frac{155}{5} = 31 \Rightarrow \boxed{\text{Mod-31 counter}}$
 $\Rightarrow$ no. of f/f's required to implement FSM:
 $2^n \geq 31$
 $\Rightarrow \underline{n = 5} \text{ f/f's. (minimum no. of f/f's required)}$

9. The output of 5-bit ripple counter is $Q_4Q_3Q_2Q_1Q_0$. The outputs $Q_4Q_2Q_1$ are connected as inputs to a NAND gate whose output is supplied as CLR input to all flipflops of the counter. Find the modulus of the counter.



When $CLR = 0 \Rightarrow Q_{n+1} = 0$

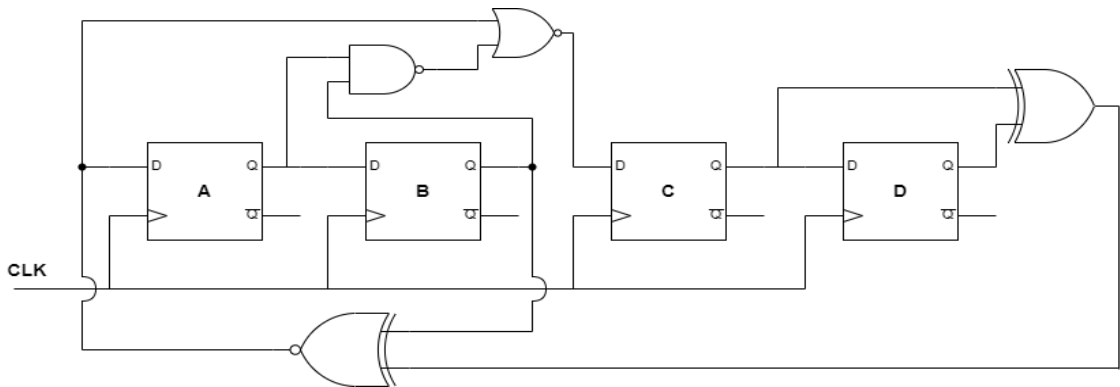


$$(1 \ 0 \ 1 \ 1 \ 0) = (22)_{10}$$

$Q_4 \ Q_3 \ Q_2 \ Q_1 \ Q_0$

$\Rightarrow$ Mod - 22 counter

10. Find the status of the given counter after 243 clock pulses with initial condition 1000.



CLK	Q_A	Q_B	Q_C	Q_D
1	1	0	0	0
2	1	1	0	0
3	0	1	1	0
4	1	0	0	1
5	0	1	0	0
6	0	0	0	0

6 clk pulses are required for the counter to get back to the initial status.

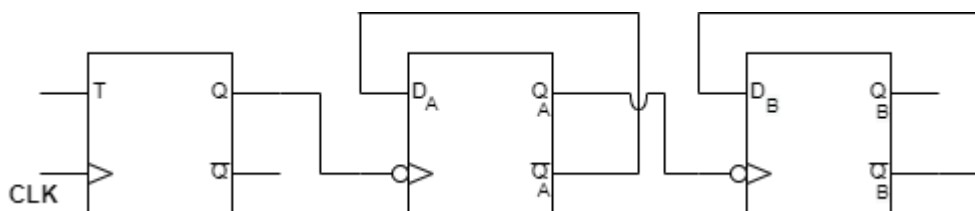
$\Rightarrow$ Mod - 6 counter

After 243 clk pulses;

$$\frac{243}{6} = 40 + 3 \text{ clk pulses}$$

the status of the counter will be 1 0 0 1.

11. Consider the given circuit with 1 T-flipflop and 2 D-flipflops whose output are Q , Q_A and Q_B . Write the counting sequence ($Q_B Q_A Q$) and comment on the behaviour of the circuit. Consider $T = 1$ always and assume initial status of all flipflops as 0.

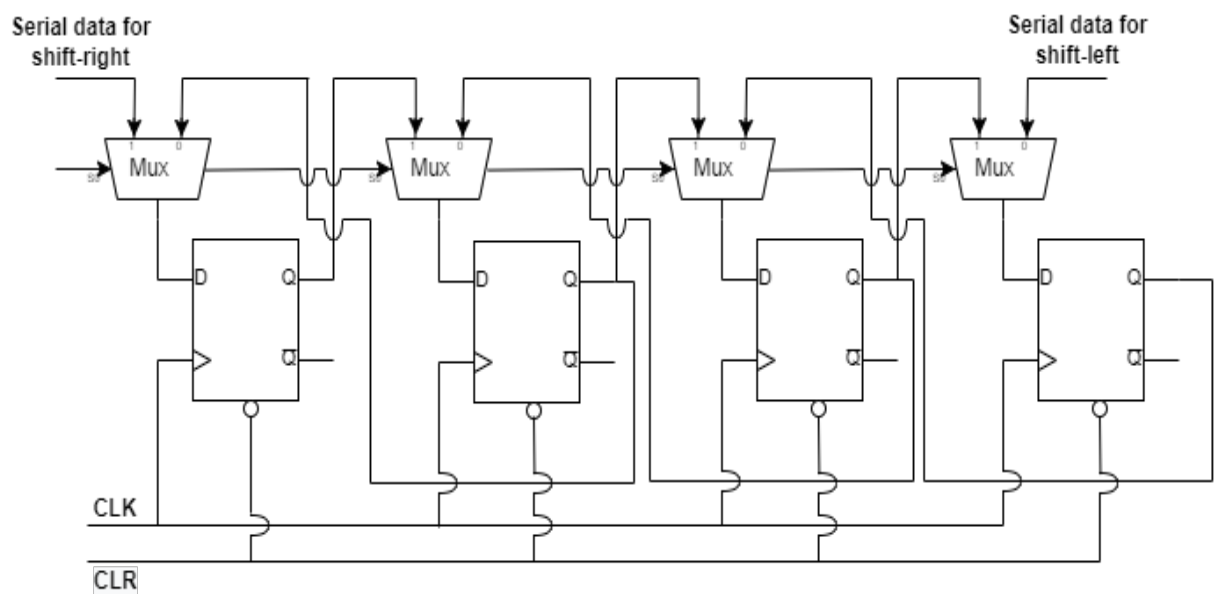


clk	Q_B	Q_A	Q
—	0	0	0
↑	0	0	1
↑	0	1	0
↑	0	1	1
↑	1	0	0
↑	1	0	1
↑	1	1	0
↑	1	1	1
↑	0	0	0

⇒ up-counting sequence
 ⇒ binary up-counter

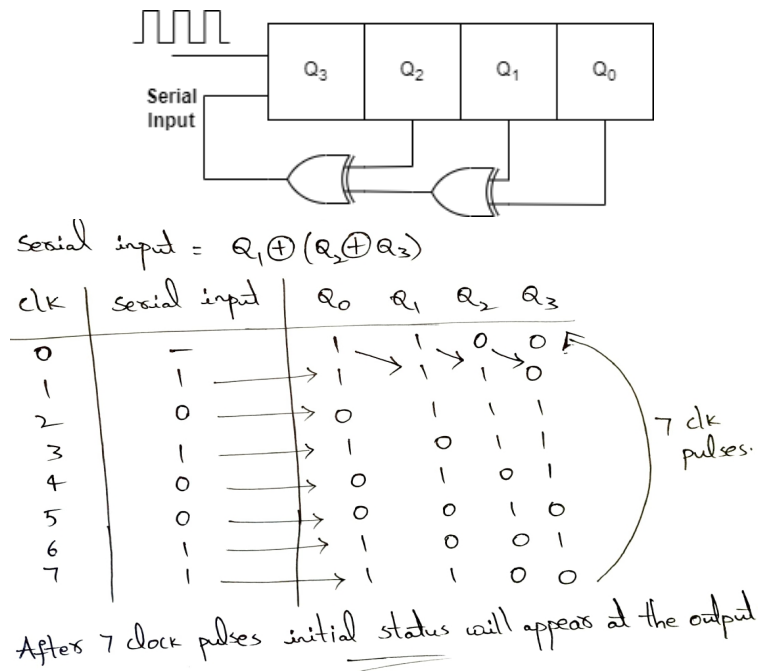
Practice Questions:

- Design a 4-bit bidirectional shift register and explain its working.

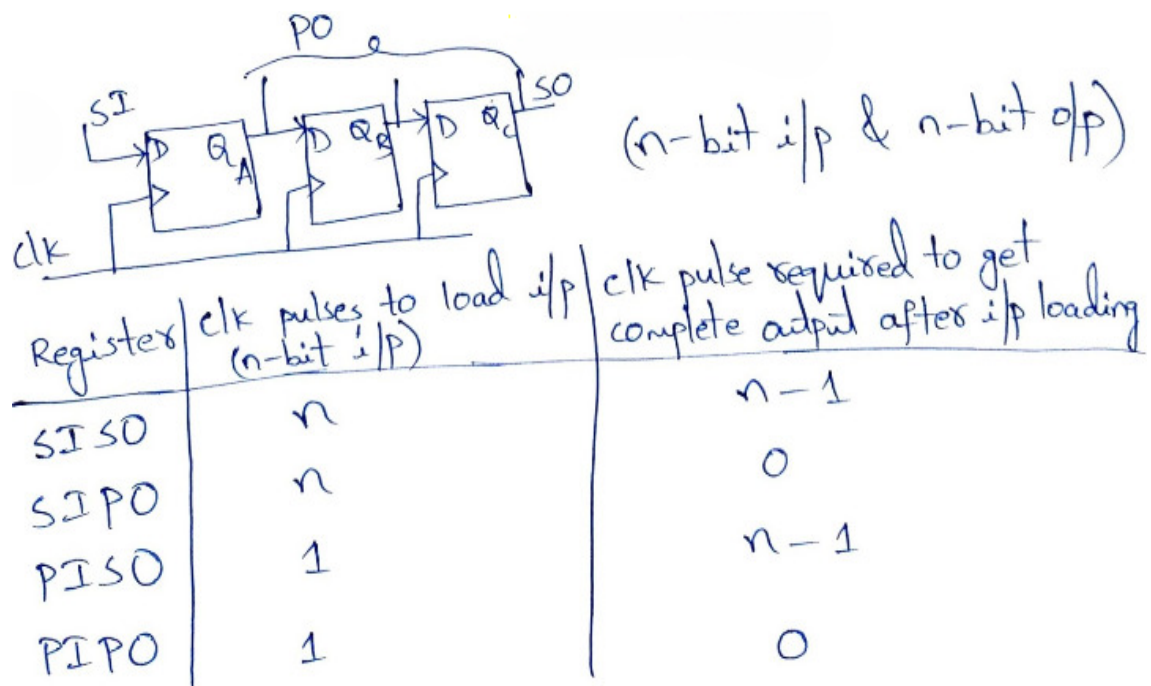


Mode control S_0	Register operation
0	shift-left
1	shift-right

- A shift register that shifts the bit one position to the right at each clock pulse is initialized to 1100 as shown in figure. After which clock cycle will initial content of the register reappear at the output?



3. List out the number of clock cycles required by n-bit SISO, SIPO, PISO and PIPO registers for completely loading n-bit inputs and observing complete outputs (after completely loading the input) respectively.



4. Find the number of flipflops required to design a mod-16 counter in terms of (i) ripple counter, (ii) twisted ring counter and (iii) ring counters.

- iv. Mod-16 counter counts 16 clk pulses.
- i) ripple counter: counts 2^n clk pulses.
 $2^n = 16 \Rightarrow n = \underline{\underline{4}}$ f/f s.
- ii) twisted ring counter: counts $2n$ clk pulses
 (Johnson) $2n = 16 \Rightarrow n = \underline{\underline{8}}$ f/f s.
- iii) ring counter: counts n clk pulses
 $n = \underline{\underline{16}}$ f/f s.

5. A binary ripple counter uses flipflops that trigger on the positive edge of the clock. What will be the counter if (a) the normal outputs of the flip-flops are connected to the clock and (b) the complement outputs of the flip-flops are connected to the clock?

Solution: (a) Down counter and (b) Up counter.

[illegible]

logic 1

